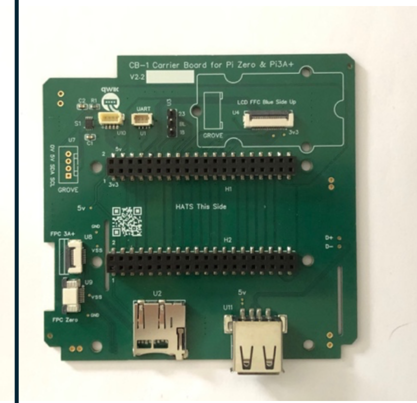
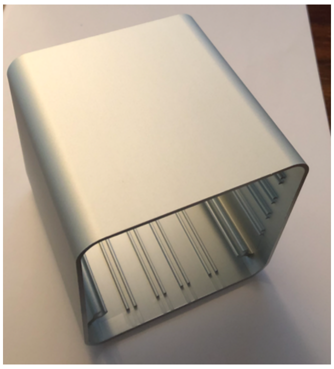
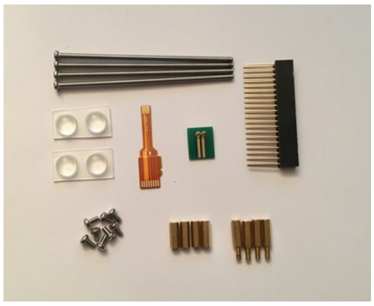
PIXIS Assembly Instructions

Raspberry Pi Zero 2W, Pixis CB-1 Carrier Board and 90EXT Case.

1. Start with these parts. Use the Raspberry Pi Imager software to prepare a microSD card with a bootable OS for step 8. Ensure you enable ssh and enter your home network Wifi SSID and password as you will need these later.   https://raspberrypi.com/software/



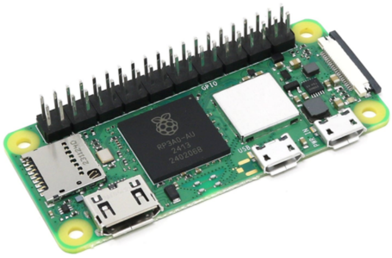




CB-1

90EXTS (or EXTB)

SSK1

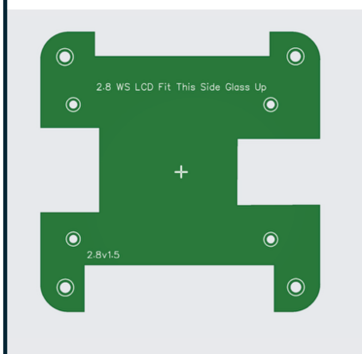


PIZ2W

BPZ

BP (If No LCD)

OPTIONAL LCD PARTS

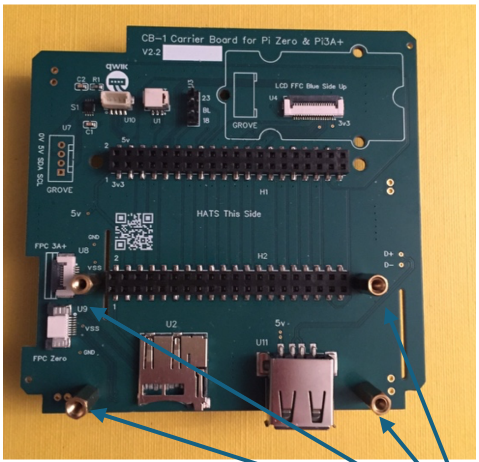


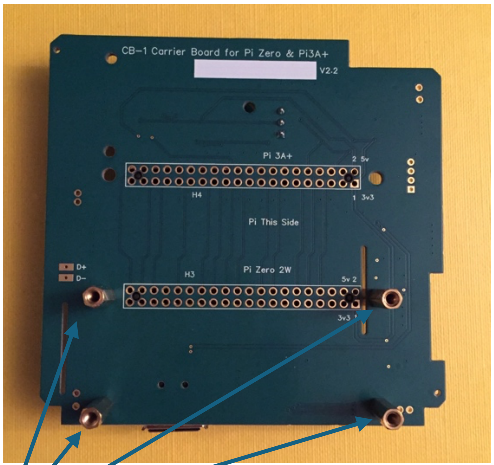
CB-3

2.8WSL or 2.8WSS

FP28

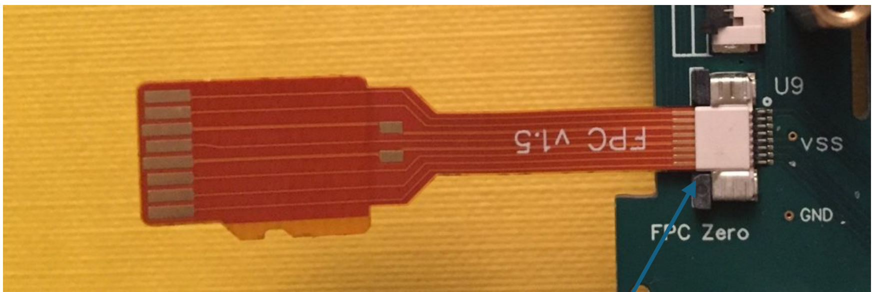
1. Fix the 8 x 12 mm spacers from the CB-1 Kit to the CB-1 board.
    a) 4 x top HAT-side spacers are 12 mm male – female.
    b) 4 x bottom Pi-side spacers are 12 mm female – female.





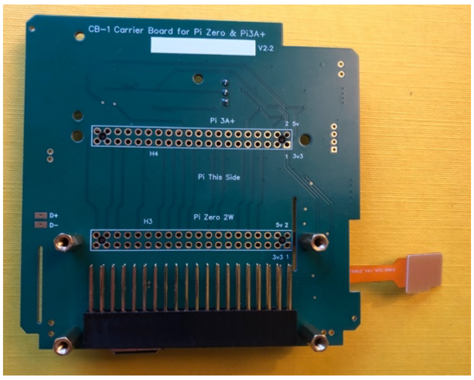
8 x 12 mm spacers

2. Fit the microSD Card flex cable to the CB-1 connector U9.
    a) Pull out the black lever of FPC connector U9.
    b) Insert the flexi cable with gold fingers facing up.
    c) Ensure the FPC gold fingers are pushed into the connector sufficiently.
    d) Push the black lever back into the connector and press in tight.

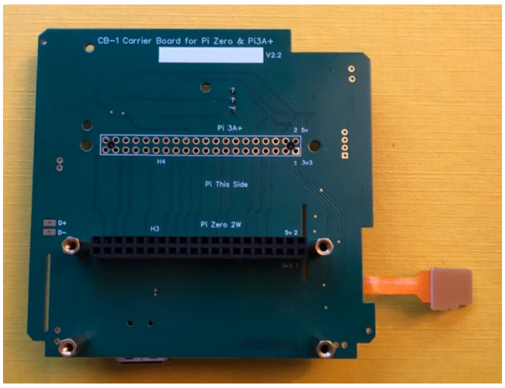


Align

3. Fit the 40-pin header to the CB-1 board pushing carefully the long pins into the connector holes marked H3 and Pi Zero 2W.

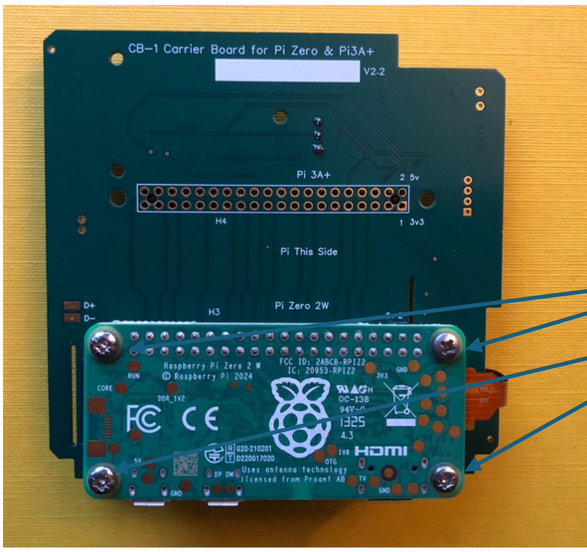


CB-1 with 40-pin Header for Zero



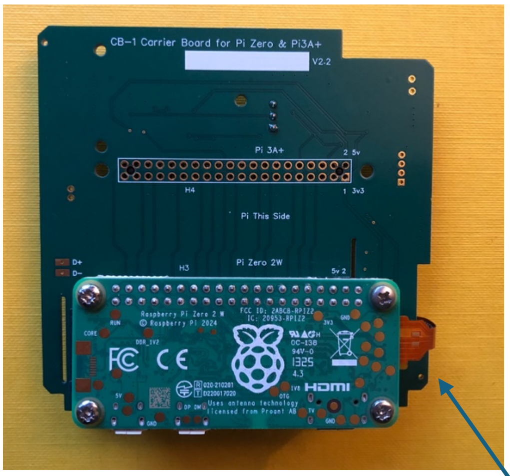
40-pin Header inserted

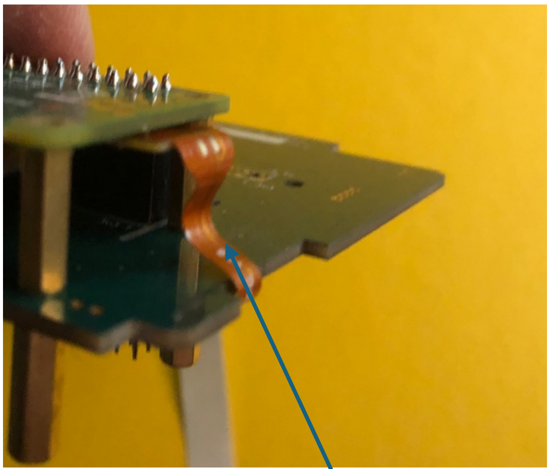
4. Fit the Pi Zero 2W to CB-1 with 4 x M2.5 screws. Note the CB-1 board has the legend “Pi This Side”.



4 x M2.5 Fixing Screws

5. Fold over and fit the microSD Card flex cable into the Pi Zero 2W microSD Card holder. Squeeze the flexi cable over the CB-1 pcb edge to ensure the flexi cable does not foul the PCB runners inside the case when the assembly is slid into the case. Bend the flexi cable inwards as shown below. Please take care and take your time to ensure a good tight fit.

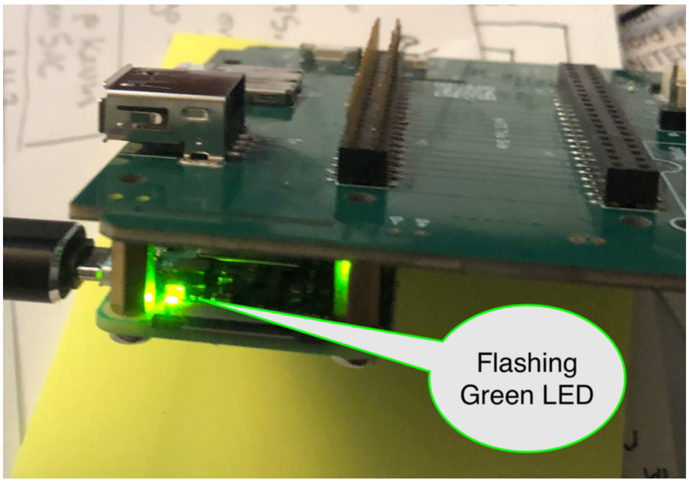




Insert SD Card stiffener

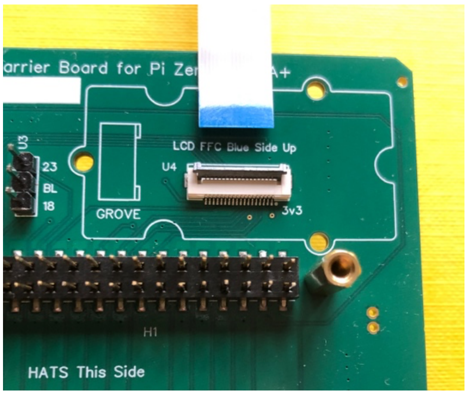
Bend Flexi circuit inwards

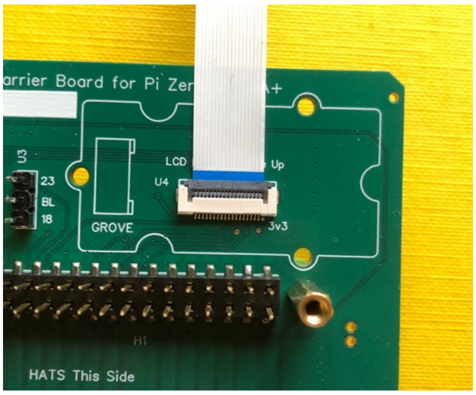
6. To ensure the cable has been inserted correctly now is a good time to fit a known-good, OS-loaded microSD card to the CB-1 board and power up the assembly to ensure the Pi Zero 2W boots OK. A flashing green activity LED on the Zero 2W in the first 30 seconds is a good sign. If there is no flashing green activity LED, power down and double check the flexi cable connections.



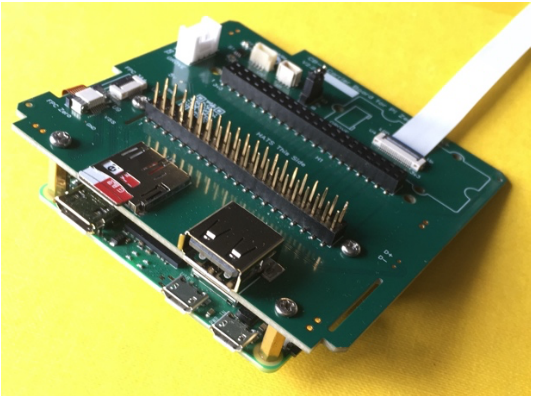
7. If you are installing a 2.8-inch SPI LCD please attach the FPCSPI LCD SPI flat cable to U4 on CB-1. Flip the black catch up, insert the FPC ensuring the blue stiffener is facing up and then close the catch down firmly. Check the jumper on U3 is BL-18. This sets the LCD backlight to the default GPIO 18. Should you need to use GPIO 18 in your project for something else, then placing the jumper at BL-23 alters the backlight enable pin to GPIO 23. Edit the config.txt file accordingly.

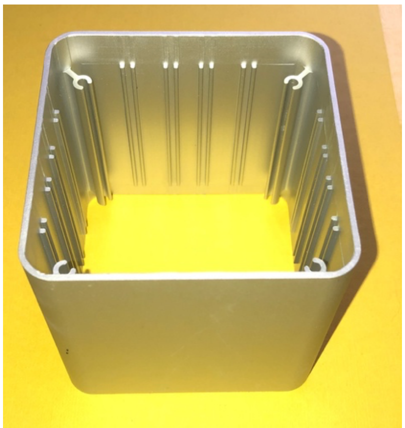
   See https://github.com/PIXISREPO/PIXIS-CODE/





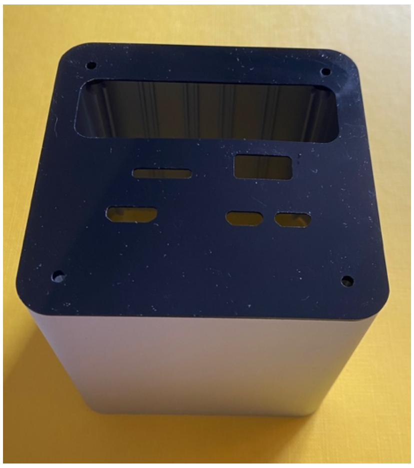
8. Your assembly should now look like this and is ready to slide into the case. As the case is a square profile and the PCB sliders are all equal it does not matter which way round you fit the assembly. What does matter is which way up the case is. The PCB sliders are machined down into the body of the case with one end deeper than the other. The shallow end is where you will mount the front panel with the optional LCD and the deeper end is where you will mount the back panel for the input and output connectors.





9. Fit the back panel to the deep end of the case (A). Make sure it is pushed in securely then turn the case over so the top is open upwards (B). Then slide the assembly into the case using the second from the top set of runners. IMPORTANT: Take great care not to let the flex cable foul the PCB runners as the assembly is slid into the case. The FPC will be on the right hand side of the case as you slide the assembly in.

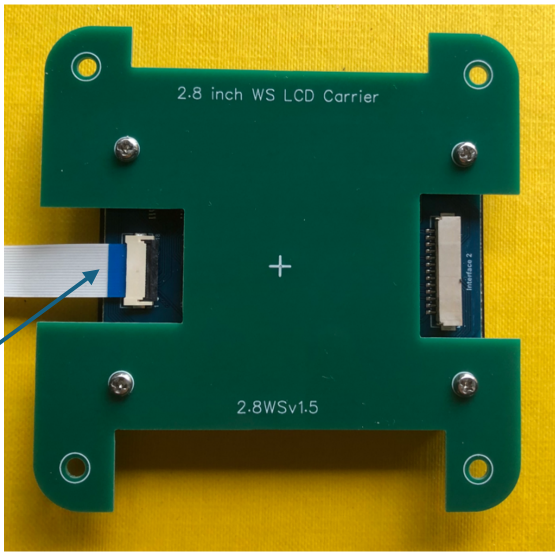
(A)



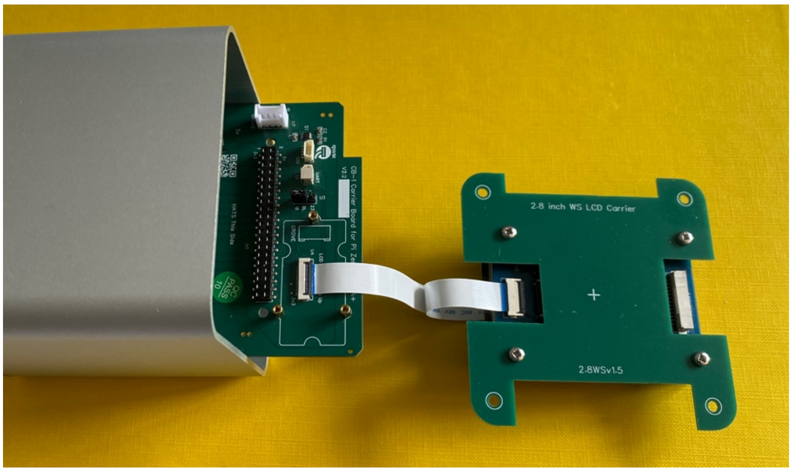
(B)

10. If you are fitting the 2.8-inch SPI LCD now is the time to assemble the LCD to the carrier board using the four M2.5 screws provided. Ensure the LCD is mounted on the correct side of the carrier board 'Glass side up'. Turn the LCD and carrier board over and connect the FFC to the LCD FFC connector. Ensure the blue side of the FFC is facing upwards and is visible after fitting. Flip the back side of the LCD FFC connector lever up – insert the blue end of the FFC carefully and then close the lever down firmly. The finished LCD sub-assembly should then look like this.

Flexi Cable Blue side up

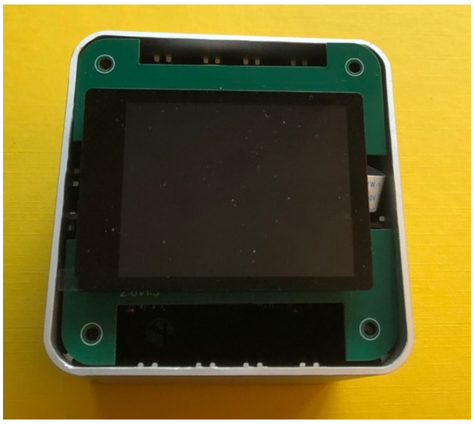


11. The other end of the LCD FFC was previously connected to U4 on CB-1 in Step 8. The blue side of the FFC should be visible at both ends after fitting.



12. Place the LCD sub-assembly into the case top. The silk screen legends on the LCD carrier board should be right way up, i.e. not upside down when the case is held upright and the Pi Zero 2W connectors are at the bottom of the case.

The front and back view of the assembled case should now look like this.



13. Remove the plastic protection from the LCD front face and the LCD Front Panel and fit the front panel onto the case over the LCD. The LCD cut out is centred so it does not matter which way round it goes. An elastic band or some masking tape is useful to temporarily hold the front and back panels in place whilst the four long fixing bolts are inserted into the case from the back. Screw each bolt down finger tight only and ensure the front and back panels are aligned with the case and that the 1 mm lips on the inside edge of the panels are inside the case. Screw down the four bolts taking care not to overtighten.

    The finished assembly should now look like this. Fit the HAT blanking panel to close the HAT connector opening and this will finish off the assembly.


2. When the assembly is complete, power up the CB-1 and install the Drivers for the Waveshare 2.8-inch SPI LCD - SKU 27579. The PIXIS display drivers and the required config.txt settings for both Landscape and Portrait orientation are available from the PIXIS-CODE GitHub repository.

   https://github.com/PIXISREPO/PIXIS-CODE/

   Follow the README instructions, download the raw binaries and copy them to the /usr/lib/firmware directory. Add the corresponding Landscape or Portrait configuration text block to the [all] section of /boot/firmware/config.txt.

   The standard PIXIS configuration uses U3 at BL-18 (backlight-gpio=18)
   If you require GPIO 18 for other purposes then move the U3 jumper
   To BL-23. Remember to change the backlight setting to backlight-gpio=23.

3. Should you wish to 3D print or CNC machine either front ot back panels to accommodate different displays or HAT connector openings then CAD and CAM files for the PIXIS case and panels are available at

   https://github.com/PIXISREPO/PIXIS-CAD/

4. A selection of ready made panel parts are available at the Pixis Store

   https://pixis.uk/products/